THE A-TEAM

HOUSING PRICE ANALYSIS



Analysing housing prices and visualising machine learning predictions

PROJECT SUMMARY

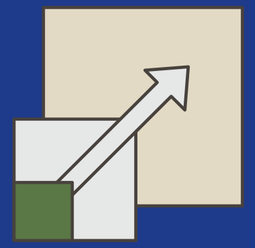
- For our capstone, THE A-TEAM (Favour, Trizah, Tanran, and Latita) conducted an end-to-end analysis of the Ames Housing dataset to predict residential property valuations and uncover the key factors driving real estate prices.

- We implemented a hybrid data workflow: extracting and cleaning data using SQL, conducting Exploratory Data Analysis (EDA) in Python, and building a Machine Learning pipeline to predict house prices.

- Finally, we consolidated our insights into an interactive Power BI dashboard for easy stakeholder reporting.

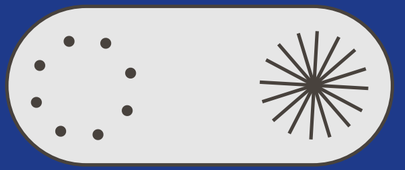


PROJECT PROBLEM



In the real estate market, inaccurately predicting property values can lead to massive financial miscalculations for buyers, sellers, and investors.

It is often difficult to quantify exactly how much value specific physical features (like square footage, overall condition, or a pool) add to a property, and how non-linear factors like neighborhood or zoning impact the final valuation. We needed a data-driven way to minimize these valuation errors.



PROPOSED SOLUTION



- We proposed building a predictive Machine Learning pipeline to accurately forecast house prices (SalePrice).

- Based on feedback from our mentor, we established a baseline using Linear Regression and then trained a Random Forest Regressor to capture complex, non-linear relationships in the data.

- We optimized our models using RMSE (Root Mean Squared Error) to heavily penalize large, costly pricing mistakes. We also built an interactive Power BI dashboard so stakeholders could easily explore pricing trends, filter by property quality, and visually assess our model's accuracy.

PROJECT SCOPE



- Our project covers a full end-to-end data pipeline

- SQL: Data extraction, cohort subsetting, and baseline metrics.
- Python (EDA): Statistical distributions, spatial/outlier analysis, handling missing data, and feature engineering.

PROJECT SCOPE



- Our project covers a full end-to-end data pipeline

- Machine Learning: Training, hyperparameter tuning, and comparing Linear Regression vs. Random Forest models, alongside extracting the top 10 most influential property features.

- Power BI: A multi-page interactive dashboard featuring market summaries, average price by neighborhood, and an actual vs. predicted model performance scatter plot.



LIMITATIONS

The scope is strictly focused on predicting current valuations and identifying feature importance. It does not forecast future market trends or predict when/where the next house will be sold, as that would require external historical economic data outside the dataset.

WHAT WE DID: PYTHON



- Data Distributions: Used Pandas and Seaborn to compute summary statistics and plot histograms to evaluate the right-skewed distribution of housing prices.
- Hypothesis Testing & Visualization: Formulated testable hypotheses, detected extreme multivariate outliers using boxplots, and engineered domain-specific variables to deepen the analysis.
- Statistical Testing Selection: Identified that heavily skewed data with extreme outliers required non-parametric statistical tests (like the Wilcoxon Signed-Rank Test) rather than traditional Paired T-Tests.

KEY FINDINGS

- Aggregation Insights: By utilizing GROUP BY functions, the team successfully generated neighborhood-level summary statistics, instantly identifying which neighborhoods had the highest volume of home sales and the highest average asking prices before any Python visuals were drawn.

- Data Quality Realization: The manual process of querying for missing values highlighted the importance of data integrity; if those NULLs had not been identified in SQL, the subsequent Machine Learning algorithms in Python would have crashed or produced skewed results.

- Pipeline Success: The SQL workflow proved that properly filtering and structuring data in a relational database dramatically streamlines the exploratory data analysis and modeling phases that follow.



WHAT WE DID:

SQL



- The analysis began with setting up an SQLite database and populating it with dataset records transferred directly from Pandas DataFrames.

- We performed schema inspections to understand property attributes and table structures, followed by applying filtering conditions to isolate Residential Low-Density properties sold under Normal sale conditions.

- Aggregate SQL queries were executed to compute housing counts and average sale prices across neighborhoods, alongside dedicated checks to pinpoint missing values in critical feature columns (e.g., Lot Frontage, Basement attributes).

KEY FINDINGS

- The tuned Random Forest significantly outperformed the baseline Linear Regression model, successfully reducing the RMSE and explaining roughly 91.67% of the price variation.

- Feature importance extraction confirmed that core structural variables—specifically Overall Quality, Above Ground Living Area, and Neighborhood—are the dominant drivers of a property's final valuation.



WHAT WE DID: MACHINE LEARNING



- Baseline Modeling: Implemented a baseline Multiple Linear Regression model using an 80/20 train-test split to establish an initial performance benchmark.
- Advanced Modeling: Trained an ensemble Decision Tree model (Random Forest Regressor), utilizing hyperparameter tuning to capture complex, non-linear relationships between property features.
- Metric Alignment: Evaluated model performance strictly using RMSE (Root Mean Squared Error) to heavily penalize large, costly real estate valuation mistakes, alongside MAE and R^2.

KEY FINDINGS

- The tuned Random Forest significantly outperformed the baseline Linear Regression model, successfully reducing the RMSE and explaining roughly 91.67% of the price variation.

- Feature importance extraction confirmed that core structural variables—specifically Overall Quality, Above Ground Living Area, and Neighborhood—are the dominant drivers of a property's final valuation.



WHAT WE DID: POWER BI



- Data Import & DAX: Imported cleansed Ames dataset and ML predictions; created measures for avg sale price, total sold, and prediction error.
- Price Analysis: Built a clustered bar chart to compare average house prices across neighbourhoods, controlled by an Overall Quality slicer.
- Model Performance: Built an actual vs. predicted scatter plot to assess prediction accuracy using ML evaluation results.

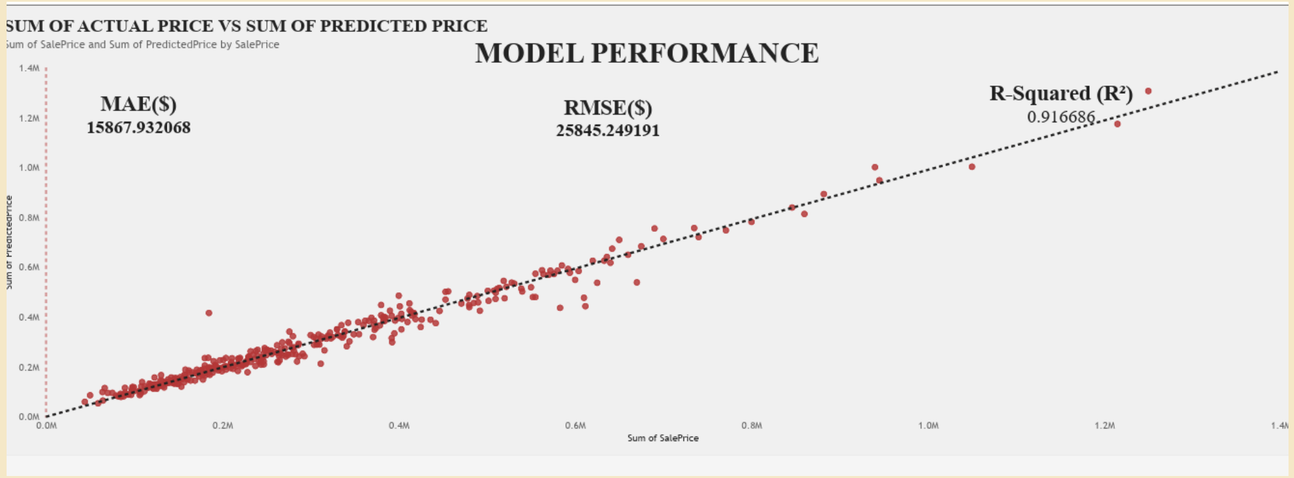
KEY FINDINGS

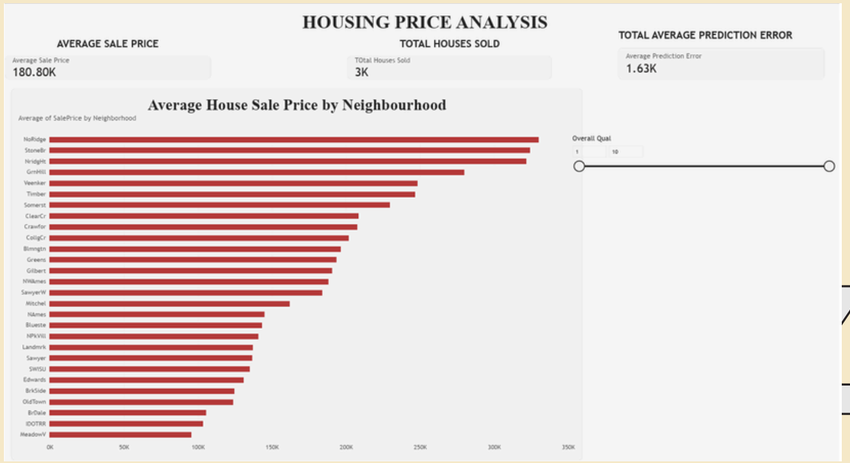
- Market Overview: Across the dataset, the Average Sale Price sits at $180.80K, with a total of 3K properties analyzed.

- Geographic Price Disparities: The Clustered Bar Chart revealed massive pricing variations strictly based on geographic location and neighborhood boundaries.

- Model Accuracy & Evaluation: The scatter plot mapping Actual vs. Predicted prices visually confirmed the strength of the Random Forest model.

- Final Metrics: The dashboard confirmed an average prediction error of just 1.63K, with the ML model successfully explaining 91.67% (R^2) of the total variance in housing prices. The RMSE of $25,845.25 highlights the model's sensitivity to extreme outliers.







CONCLUSION

The Ames Housing Capstone project successfully demonstrated the effectiveness of integrating modern data analytics tools to solve complex valuation challenges. By seamlessly combining SQL, Python, Machine Learning, and Power BI, the team built a robust end-to-end data pipeline that significantly minimized real estate pricing errors.

The analytical workflow effectively identified critical structural and geographic drivers influencing property prices, delivering actionable market insights. Overall, this comprehensive approach highlights the invaluable role of an end-to-end data strategy in enhancing predictive accuracy and supporting data-driven real estate decisions.

INTRODUCING THE A-TEAM



TRIZAH NASUMBA

SQL ANALYST



TANRAN ETHRUDIES SUHLIY

POWER BI DEVELOPER



FAVOUR CHIBAYA

PYTHON ANALYST



LATITA MVUNELO

TEAM LEADER AND ML LEAD

A-TEAM

THANK YOU

